

---

# Redesigning Humanized Monoclonal Antibodies Through Computational Approaches

---

Tina Zhang<sup>1</sup> Zichen Deng<sup>2</sup> Peter Lin<sup>2</sup> Edward Li<sup>1</sup>

## Abstract

Therapeutic antibody design is constrained by costly in vivo validation pipelines. We present a four-phase computational workflow integrating fine-tuned protein language models with structure-based generative design for target-specific antibody optimization. We first benchmark fitness predictors on the FLAb suite, finding that frozen ESM-2 (650M) with ridge regression achieves Spearman correlations of 0.47, 0.73, and 0.48 on Koenig binding, Koenig expression, and Warszawski binding tasks. Fine-tuning ESM-2 on these tasks improves Spearman by 0.056–0.080 across all three benchmarks. Using this surrogate as a scorer, we apply RFdiffusion backbone generation and ProteinMPNN sequence design to produce HER2-targeting antibody candidates, reranked by Pareto optimization over predicted binding and developability. Our end-to-end pipeline demonstrates a promising approach for computationally guided antibody redesign with reduced reliance on wet-lab evaluation.

## 1. Introduction

Therapeutic antibodies are essential in modern oncology, with essential examples such as trastuzumab (HER2-targeting) and cetuximab (EGFR-targeting) demonstrating sustained clinical impact (Toledo-Stuardo et al., 2024). However, despite this success, the development pipeline remains heavily dependent on animal immunogenicity studies and in vivo validation experiments, which creates major bottlenecks for large-scale candidate screening (Wang et al., 2025).

Recent advances in protein foundation models and diffusion-

based structural design offer a potential step toward in silico-accelerated antibody engineering. ESM-2 (Lin et al., 2023) and ProGen3 have demonstrated state-of-the-art performance on protein structure and function prediction, while RFdiffusion (Bennett et al., 2026) enables de novo backbone generation with competitive binding geometry. Despite these advances, key developability properties, including solubility, flexibility, and functional activity (e.g., ADCC, immunogenicity), remain insufficiently captured by current models (Park et al., 2025; Niekerk et al., 2026).

We propose a multi-phase workflow: (0) benchmark fitness predictors on FLAb to select a surrogate scorer, (1) fine-tune ESM-2 for improved antibody fitness prediction, (2) generate HER2-targeting candidates via RFdiffusion and ProteinMPNN, and (3) rerank candidates by Pareto optimality over predicted binding and developability. We evaluate top reranked candidates against a random baseline using ipTM and pTM scores.

## 2. Methods

### 2.1. Datasets

We evaluate on three FLAb benchmark datasets (Chungyoun & Gray, 2025): (1) *Koenig binding (G6)*, which measures binding affinity of anti-VEGF antibody variants via phage display (Koenig et al., 2017); (2) *Koenig expression (G6)*, which measures functional expression levels of the same variants via protein A/L binding; and (3) *Warszawski binding (D44)*, which maps the mutational landscape of a variable domain of an anti-lysozyme antibody via deep mutational scanning (Warszawski et al., 2019). All tasks are regression problems evaluated using 10-fold cross-validation random splits for the datasets consists of interpolated mutations on a single antibody scaffold.

### 2.2. Phase 0: Baseline Fitness Predictors

We compare five methods as fitness predictors. (1) *Mean baseline*: predicts the training mean for all test samples. (2) *Sequence with ridge*: counts per-residue amino acid frequencies in heavy and light chains and fits ridge regression ( $\alpha = 1.0$ ), which tests whether composition alone carries predictive signal. (3) *ESM-2 with ridge*: extracts frozen

---

<sup>1</sup>Department of EECS, Massachusetts Institute of Technology, Cambridge, MA, USA <sup>2</sup>Department of Computational Biology, Harvard University, Cambridge, MA, USA. Correspondence to: Tina Zhang <zhangtin@mit.edu>.

embeddings from ESM-2 (650M) and fits ridge regression, which tests whether broad protein pretraining transfers to antibody fitness. (4) *AbMAP with ridge*: substitutes AbMAP embeddings, fine-tuned for antibody hypervariable regions, to isolate the benefit of domain-specific pretraining. (5) *ESM-2 with MLP*: replaces ridge regression with a multi-layer perceptron over frozen ESM-2 embeddings, allowing us to test nonlinear readout capacity. We compute Spearman, Pearson, root mean square error (RMSE), mean average error (MAE), and  $R^2$  averaged across held-out folds.

### 2.3. Phase 1: Fine-Tuning ESM-2

To improve upon frozen ESM-2 embeddings, we fine-tune ESM-2 on the FLAb tasks. We perform hyperparameter sweeps over backbone learning rates  $\text{lr} \in \{10^{-4}, 10^{-5}, 10^{-6}\}$  and layer-unfreezing depth  $d \in \{0, 2, 4\}$ , which is the number of unfrozen transformer layers from the top, on ESM2-35M to identify optimal configurations efficiently. The best configuration across all three tasks is  $\text{lr} = 10^{-4}$ ,  $d = 4$ , achieving Spearman correlations of 0.399, 0.785, and 0.406 on Koenig binding, Koenig expression, and Warszawski binding respectively on the 35M parameter model. This configuration is then transferred to ESM2-650M (650M parameters) and evaluated using the same 10-fold protocol, finally yielding the final fine-tuned surrogate. Model checkpoints are uploaded to HuggingFace for API access and checkpoint ensemble scoring.

### 2.4. Phase 2: Generative Antibody Design

We generate HER2-conditioned antibody candidates using RFdiffusion, initializing from the HER2–trastuzumab complex structure (RCSB PDB: 1N8Z) and the scFv scaffold (hu-4D5-8-Fv.pdb). Five backbone structures are sampled under deterministic conditions, with binding guided by three hotspot residues (T30, T31, T32) defined as target residues within 4.5 Å of the original trastuzumab interface. Full CDR length constraints are applied during backbone generation. For each backbone, ProteinMPNN is used to design 8 sequence variants with a sampling temperature of 0.2.

Separately, we run a Germinal-based generative pipeline for PD-L1, which produced handoff CSVs used for reranking. For HER2, the Germinal run did not pass quality filters, so we proceeded with RFdiffusion-generated candidates.

### 2.5. Phase 3: Surrogate Scoring and Reranking

Generated candidates are scored using the fine-tuned ESM2-650M model from Phase 1. For each candidate, we compute z-score for each binding and expression tasks, of which are derived from the respective fine-tuned model outputs. Candidates are reranked by composite score of two bindings and one expression tasks with equal weight to perform

multi-objectives Pareto front selection. Top candidates were retained for downstream validation.

### 2.6. Phase 4: Structural Validation

The top 5 Pareto-optimal HER2 candidates and 5 other random chosen candidates are submitted to Chai-1 for structure prediction and evaluation of ipTM and pTM scores, which provides an independent structural validation of the reranked candidates.

## 3. Results

### 3.1. Baseline Fitness Prediction

Table 1 reports baseline benchmarking results. ESM-2 (650M parameters) with ridge regression significantly outperforms the mean and sequence composition baselines across all three tasks, achieving Spearman correlations of 0.47, 0.73, and 0.48 on Koenig binding, Koenig expression, and Warszawski binding respectively, compared to near zero for the mean baseline. The expression task shows the largest improvement, which suggests that expression-level fitness may be more linearly encoded in ESM-2’s latent space than binding affinity.

Notably, AbMAP underperforms ESM-2 on all three tasks, with a notable gap on Koenig expression (Spearman 0.729 vs. 0.544), which implies that antibody-specific fine-tuning may hurt generalization here, possibly due to distribution mismatch with the D44 antibody. Furthermore, ESM-2 with ridge matches or outperforms the MLP on Spearman across all tasks. Overall, ESM-2 with ridge regression is the strongest and most consistent predictor, leading us to select it as our Phase 1 fine-tuning starting point.

Table 1. Baseline benchmarking on FLAb tasks (10-fold cross validation). We report Spearman and Pearson scores as primary metrics.

| Task               | Method         | Spearman     | Pearson      |
|--------------------|----------------|--------------|--------------|
| Koenig binding     | Mean           | 0.000        | 0.000        |
| Koenig binding     | Sequence+ridge | 0.174        | 0.181        |
| Koenig binding     | ESM-2+ridge    | <b>0.466</b> | <b>0.486</b> |
| Koenig binding     | AbMAP+ridge    | 0.334        | 0.361        |
| Koenig binding     | ESM-2+MLP      | 0.389        | 0.410        |
| Koenig expression  | Mean           | 0.000        | 0.000        |
| Koenig expression  | Sequence+ridge | 0.247        | 0.269        |
| Koenig expression  | ESM-2+ridge    | <b>0.729</b> | 0.746        |
| Koenig expression  | AbMAP+ridge    | 0.544        | 0.557        |
| Koenig expression  | ESM-2+MLP      | 0.727        | <b>0.766</b> |
| Warszawski binding | Mean           | 0.000        | 0.000        |
| Warszawski binding | Sequence+ridge | 0.260        | 0.246        |
| Warszawski binding | ESM-2+ridge    | <b>0.479</b> | <b>0.481</b> |
| Warszawski binding | AbMAP+ridge    | 0.345        | 0.349        |
| Warszawski binding | ESM-2+MLP      | 0.390        | 0.398        |

### 3.2. ESM-2 Fine-Tuning Improves Fitness Prediction

Fine-tuning ESM2-650M on the FLAb tasks yields consistent improvements over the frozen model across all three benchmarks. As shown in Table 2, Spearman correlation improves from 0.466 to 0.522 (+0.056) on Koenig binding, from 0.729 to 0.809 (+0.080) on Koenig expression, and from 0.479 to 0.555 (+0.076) on Warszawski binding. These gains confirm that task-specific fine-tuning meaningfully improves surrogate model quality, with the largest benefit observed on the expression task.

Table 2. Fine-tuned ESM2-650M (backbone lr =  $10^{-4}$ , number of unfrozen transformer layers from the top  $d = 4$ ) vs. frozen baseline.

| Task              | Spearman | Pearson | $\Delta$ Spearman |
|-------------------|----------|---------|-------------------|
| Koenig binding    | 0.522    | 0.476   | +0.056            |
| Koenig expression | 0.809    | 0.836   | +0.080            |
| Warsawski binding | 0.555    | 0.529   | +0.076            |

Table 3. ESM2-35M hyperparameter sweep. We use Spearman as the primary metric to determine that the best configuration is backbone lr =  $10^{-4}$ ,  $d = 4$  across all tasks.

| lr        | $d$ | Task              | Spearman     |
|-----------|-----|-------------------|--------------|
| $10^{-4}$ | 4   | Koenig binding    | <b>0.399</b> |
| $10^{-5}$ | 4   | Koenig binding    | 0.380        |
| $10^{-4}$ | 2   | Koenig binding    | 0.378        |
| $10^{-4}$ | 0   | Koenig binding    | 0.276        |
| $10^{-4}$ | 4   | Koenig expression | <b>0.785</b> |
| $10^{-4}$ | 2   | Koenig expression | 0.779        |
| $10^{-5}$ | 4   | Koenig expression | 0.722        |
| $10^{-4}$ | 0   | Koenig expression | 0.585        |
| $10^{-4}$ | 4   | Warsawski binding | <b>0.406</b> |
| $10^{-4}$ | 2   | Warsawski binding | 0.386        |
| $10^{-5}$ | 4   | Warsawski binding | 0.317        |
| $10^{-4}$ | 0   | Warsawski binding | 0.263        |

### 3.3. HER2 Candidate Generation and Pareto Reranking

We pursue two complementary generative strategies for HER2-targeting antibody design (Figure 1). First, we run a Germinal-based pipeline seeded from the trastuzumab scFv scaffold against the HER2 crystal structure (PDB: 1N8Z), which successfully generates a docked complex but does not pass downstream quality filters (Figure 1a). Second, we use an known scFv scaffold and docked against HER2 on PyDockWEB which binding contact may be further optimized on PyDockWEB (Figure 1b). We use RFdiffusion to generate 5 HER2 structural backbones conditioning on three hot-spot residues, from which ProteinMPNN designs antibody sequences conditioned on each backbone within the CDR region. The RFdiffusion/ProteinMPNN route produces 40 candidates that proceed to Phase 3 scoring.

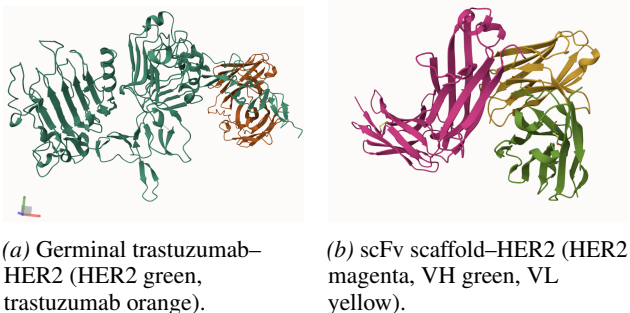


Figure 1. Phase 2 generative antibody structures visualized in Mol\*. Left: Germinal-generated trastuzumab scFv-HER2 complex using the 1N8Z template. Right: scFv antibody scaffold used for RFantibody case study docking to HER2 via PyDockWEB.

Each candidate is scored by the fine-tuned ESM-2 surrogate for both binding affinity and expression/developability fitness. Figure 2 shows the resulting Pareto front. The 10 Pareto-optimal candidates (and highlighted in orange) represent antibodies predicted to achieve the best tradeoffs between binding and developability, and are passed to Chai-1 for structural validation. Next, we examine the 3D distribution of predicted scores of three tasks (Figure 3) and reveals top-ranked candidates clustering in regions of jointly high binding and moderate expression. This suggests that the composite ESM-2 scoring effectively selects sequences with balanced performance across multiple objectives.

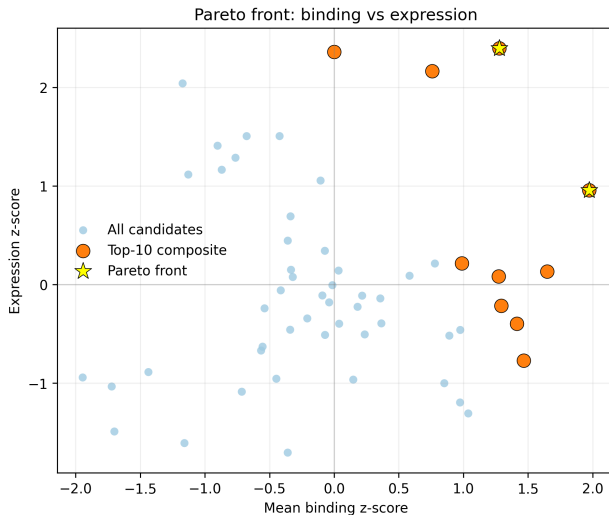


Figure 2. HER2 Pareto front. Each blue point is a generated antibody candidate scored by the fine-tuned ESM-2 surrogate. Yellow star indicates the Pareto-optimal candidates, each maximizing either mean binding and expression z-score without being dominated on the other objective.

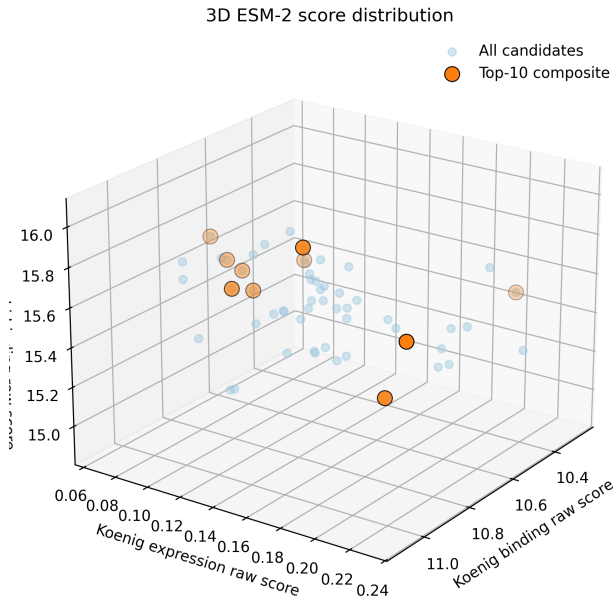


Figure 3. HER2 3D raw score. Each point represents Koenig binding, Koenig expression, and Warszawski binding raw scores of the candidates. Orange dots highlights the top candidates from the ESM2 finetuned model scoring

### 3.4. Structural Validation via Chai-1

We perform Chai-1 structural validation of the top 5 Pareto-optimal HER2 candidates selected by ESM-2 reranking and 5 random candidates not from the top 10 list. As illustrated in Figure 4, the ESM-2 top candidates achieve a mean ipTM of 0.610 and mean pTM of 0.733 compared to 0.601 and 0.722 for randomly sampled baseline candidates, which suggests stronger predicted binding interface geometry in the ESM-2-guided designs. As visualized in Figure 5, the top-ranked Chai-1 predicted complex shows the antibody Fv region docking against the HER2 extracellular domain.

Due to small sample size  $n = 5$  per group, the differences between the two groups are not significant (Welch’s t-test  $p > 0.05$ ). However, the trend demonstrates that ESM-2 reranking tends to lead to higher confidence in protein structure prediction quality and motivates validation on larger sample sizes for future work.

## 4. Discussion

Our results establish that frozen ESM-2 embeddings provide a strong, task-agnostic baseline for antibody fitness prediction, and that fine-tuning on task-specific data yields meaningful and consistent improvements. The consistent underperformance of AbMAP relative to ESM-2 suggests that domain-specific pretraining does not always confer a generalization advantage, particularly when the fine-tuning distribution diverges from the evaluation antibody scaffold

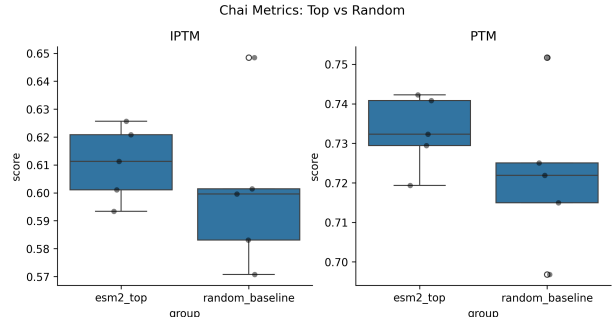


Figure 4. Chai metrics between ESM-2-selected candidates and a random baseline. ESM-2-selected candidates achieve higher mean ipTM and mean pTM scores compared to a random baseline ( $n = 5$  per group).

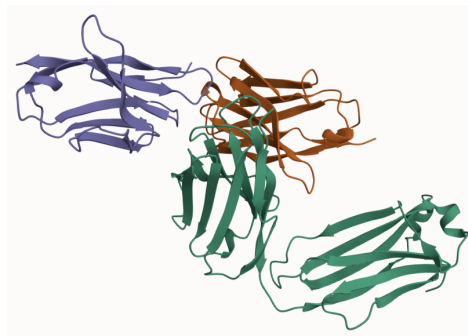


Figure 5. Chai-1 predicted structure of a top ESM-2-reranked HER2 antibody candidate (pred.model\_idx\_0). Green: HER2 target (Entity A); orange: VH chain (Entity B); purple: VL chain (Entity C). The antibody Fv region docks against the HER2 extracellular domain, consistent with the elevated ipTM scores of ESM-2-selected candidates.

(as with the D44 system). This emphasizes the importance of careful alignment between surrogate model pretraining and target antibody families.

A key limitation of the current workflow is the difficulty in generating high-quality HER2 candidates through the Germinal pipeline. RFantibody with ProteinMPNN provided a more robust generative alternative but introduces a structural prior that may not fully capture sequence-level diversity. Additionally, our surrogate scoring relies solely on ESM-2 embeddings, which do not incorporate explicit structural context at inference time. Integrating structure-conditioned models as scorers (e.g., ESMFold-derived representations) may further improve reranking fidelity.

Future work includes extending the pipeline to additional therapeutic targets and exploring ensemble-based surrogate scoring using the uploaded HuggingFace model checkpoints, expanding on multi-objective optimization on developability metrics, and finetuning of the RFantibody pipeline using classifier guided diffusion.

## Conclusion

We present a modular, end-to-end computational workflow for antibody redesign that combines fine-tuned protein language model scoring with structure-based generative design. Fine-tuning ESM-2 on FLAb tasks consistently improves Spearman correlation by 0.056–0.080, and Pareto reranking over the fine-tuned surrogate selects a focused set of HER2 candidates for structural validation, increasing mean ipTM and pTM by 0.009 and 0.011, respectively. This pipeline offers a scalable alternative to in vivo screening for early-stage antibody candidate prioritization.

## Author Contributions

Authors worked together on Phase 0 baseline benchmarking. Tina Zhang led Phase 1 ESM-2 fine-tuning and model checkpoint management. Zichen Deng led Phase 2 generative pipeline design, including Germinal and RFDiffusion/ProteinMPNN. Peter Lin led the pipeline design and Phase 3 surrogate scoring and Pareto reranking. Edward Li led Phase 4 Chai-1 structural validation.

## References

- Bennett, N. R., Watson, J. L., Ragotte, R. J., et al. Atomically accurate de novo design of antibodies with RFDiffusion. *Nature*, 649(8095):183–193, 2026. doi: 10.1038/s41586-025-09721-5.
- Chungyoun, M. and Gray, J. Fitness landscape for antibodies 2: Benchmarking reveals that protein AI models cannot yet consistently predict developability properties. *bioRxiv*, 2025. doi: 10.64898/2025.12.27.696706.
- Koenig, P., Lee, C. V., Walters, B. T., Janakiraman, V., Stinson, J., Patapoff, T. W., and Fuh, G. Mutational landscape of antibody variable domains reveals a switch modulating the interdomain conformational dynamics and antigen binding. *Proceedings of the National Academy of Sciences*, 114(4):E486–E495, 2017. doi: 10.1073/pnas.1613231114.
- Lin, Z. et al. Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science*, 379(6637):1123–1130, 2023. doi: 10.1126/science.ade2574.
- Niekerk, L., Moller, J., Ritter, S., et al. 2025 ginkgo datapoints antibody developability competition outcomes: limited model performance and a call for data standardization. *MAbs*, 18(1), 2026. doi: 10.1080/19420862.2026.2634216.
- Park, S., Jeong, W., Kim, Y., Lee, C.-H., and Seok, C. Artificial intelligence in therapeutic antibody design: Advances and future prospects. *Current Opinion in Structural Biology*, 94:103084, 2025. doi: 10.1016/j.sbi.2025.103084.
- Toledo-Stuardo, K. et al. Therapeutic antibodies in oncology. *Journal of Translational Medicine*, 2024. URL <https://pmc.ncbi.nlm.nih.gov/articles/PMC11448508/>.
- Wang, X.-D., Ma, B.-Y., Lai, S.-Y., Cai, X.-J., Cong, Y.-G., Xu, J.-F., and Zhang, P.-F. High-throughput strategies for monoclonal antibody screening: Advances and challenges. *Journal of Biological Engineering*, 19:41, 2025. doi: 10.1186/s13036-025-00513-z.
- Warszawski, S., Borenstein Katz, A., Lipsh, R., et al. Optimizing antibody affinity and stability by the automated design of the variable light-heavy chain interfaces. *PLOS Computational Biology*, 15(8):e1007207, 2019. doi: 10.1371/journal.pcbi.1007207.